DHRUV KALE

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EDUCATION

Master of Science in Computer Science, RPTU Kaiserslautern — Oct 2024 - Sept 2027 (expected)
Bachelor of Engineering in Computer Engineering, University of Pune - 1.9 — 2017 - 2021

SKILLS

Programming	Python, SQL, Java (Core)
AI & LLM	Agentic AI, LangGraph, MCP, QLoRA fine-tuning, PyTorch
Engineering & Tooling	Playwright, Docker, n8n, Git, Jira
Enterprise Data	Power BI, AppDynamics, Flask
Languages Spoken	English (C2) · German (B1)

PROJECTS

- **LLM Browser Agent for Heterogeneous Enterprise Systems** *(M.Sc. Project, RPTU Kaiserslautern, 2026):* Built an LLM browser agent that operates a custom Leave Management System and an unmodified Nextcloud Calendar through their web interfaces via Playwright, exposing 15 tools over MCP, and reduced context per call from thousands of markup tokens to tens by extracting typed values from test-id nodes.

  Enforced 25 correctness guarantees in deterministic code at zero inference cost, backed by 304 tests and a structured trace of every run, and gated all actions behind user authentication for accountability.
- **LLM Fine-Tuning for Language Translation:** Fine-tuned Qwen2.5-1.5B for German-to-French translation using QLoRA, achieving an 84% increase in BLEU score through data cleaning, prompt engineering and synthetic data generation. GitHub
- **Automated Job Application Tracking Pipeline (n8n & AI):** Built an n8n workflow that ingests unstructured emails and uses an LLM to classify job application statuses, with a global error-handling framework streaming diagnostics to a Discord bot. GitHub

EXPERIENCE

Deputy Shift Lead (Operations, Working Student) — March 2025 - Present
Flink — *Kaiserslautern, Germany*

- Oversee hub operations, inventory and KPI tracking, coordinating ops and rider teams of up to 10 with a structured handover at the end of every shift.
- Manage pre- and post-order feedback, resolve rider issues in real time, and identify bottlenecks to keep operations running smoothly at an average daily order volume of up to 280.

Senior Analyst / Software Engineer (Power BI Developer) — April 2023 - Sept 2024
Capgemini — *Pune, India*

- Awarded the **Customer Delight Award** for pitching and leading the transition from manual to automated **Power BI** reporting across three weekly reports, saving roughly 6 hours of manual data compilation per week.
- Integrated data sources and developed reports using **DAX** to deliver actionable insights for business decision-making; leveraged LLMs to optimise queries for performance.
- Managed workspace access, report ownership and refreshes, and trained team members on maintenance.

Analyst / Software Engineer (Performance Engineering in Automation) Sept 2021 - March 2023
Capgemini *Pune, India*

- Developed **Python** scripts to automate generation of health status dashboards, saving 2 hours of manual configuration per cycle, and enabled automated **AppDynamics** .NET agent deployments using Python and **Ansible**.
- Built client AppDynamics dashboards monitoring application health and infrastructure metrics, and configured health rules and policies integrated with Opsgenie alerts to strengthen on-call support for L2/L3 teams.
- Delivered training sessions enabling support teams to leverage AppDynamics for monitoring and troubleshooting.

ACADEMIC SEMINARS

- **Edge Agentic Intelligence: Architecture, Memory, Protocols and Trust for Agentic AI on Edge Devices** *(Summer Semester 2026, RPTU Kaiserslautern/DFKI):* Authored a research survey on deploying autonomous, tool-using LLM agents on edge hardware. Analysed memory-level bottlenecks (KV-cache inflation, unified-memory constraints), the model-compaction-versus-reasoning-stability trade-off, multi-agent communication protocols (MCP, A2A), and confidential-computing approaches for securing edge-resident agents.
- **World Creation from Egocentric Devices** *(Winter Semester 2025/26, RPTU Kaiserslautern/DFKI):* Authored a research survey on real-time 3D scene reconstruction from smart-glasses data, covering 3D Gaussian Splatting for photorealistic rendering, Visual-Inertial Bundle Adjustment for motion-robust reconstruction, and SceneScript's structured-language scene representation with human-in-the-loop correction.

HOBBIES AND INTERESTS

Bouldering · Dance · Power Lifting · Arm Wrestling